Customer Name : M/s. Nihva Technologies

Proposal Details: **Techno Commercial Proposal for 10.8 mtr Telescopic Conveyor**

Proposal No : FRAPL/EN/19-20/218

Proposal Date : 28th Nov 2019

Revision No : R0

Cover Letter

Kind Attention **: Mr. Amit**

Dear Sir,

Greetings from **M/s. Frontier Robotics & Automation Pvt Ltd.**

We are very much Thankful to you for sharing your valuable enquiry with us.

With reference to the same we had gone through the details shared by you and we hereby propose the Best Techno-Commercial Offer as enclosed.

We hope the enclosed offer is in line with your requirement.

Awaiting your valuable feedback regarding the same.

We admire for the excellent & Healthy Long term Relations with you & your esteemed Organisation!!!

With regards,

Vipul Kasar
Director
Cell No: +91 – 74100 15000

TELESCOPIC CONVEYOR-RFQ

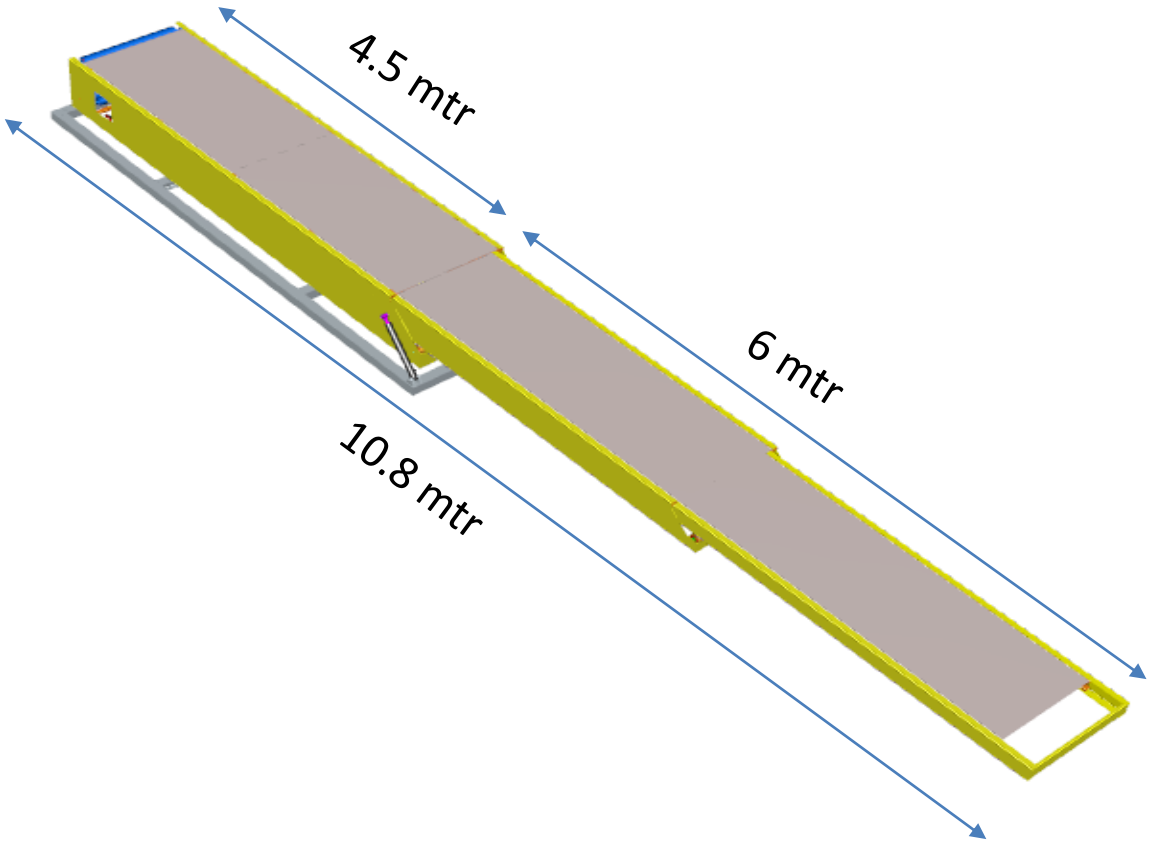


RFQ Specifications :

- ✓ Box Size: 600*300*400mm,
- ✓ Conveyor Speed: 8 mtr/min required (30 mtr/min)
- ✓ Boom Drive Speed :10-15 mtr/min
- ✓ Capacity: 167kg/mtr (1803 kg in total load)
- ✓ Belt Width : 600 mm
- ✓ Parking Space: 4.8 mtr
- ✓ Extension : 6 mtr
- ✓ Total length : 10.8 mtr

Particular	Specification
Material to be handle	Boxes/sacks
Weight of Box / bags	50 Kg Maximum
Size of Box / Bag	600 (L) x 300 (W) x 400 (H) (Overall)
Temperature	Ambient
Capacity	750 Boxes/Hour continue Operation
Type	Collapsible Boom Type Hinged Telescopic Belt Conveyor
Size of Belt	600 mm width
Type of Belt	Smooth top 2 Ply , PVC Coated Endless type (Food Grade)
Length of Conveyor	Fixed Length - 4800 mm Telescopic (Expanded) - 6000 mm Total - 10800 mm
Main Body of Conveyor	Manufacturing From 6 mm Thick IS 226 Plate Material (as per new advance design)
Height Adjustment	With 2 HP Power Pack and Cylinder for Height Adjustment (Rexroth parts) (ABB Motor)
Panel (PLC) With HMI	The Conveyor comes with a fixed main panel on rear side and front side only Operating Panel with emergency stopper sensor Bar , LED Lighting , Totally consoled wiring
Painting	Powder Coated

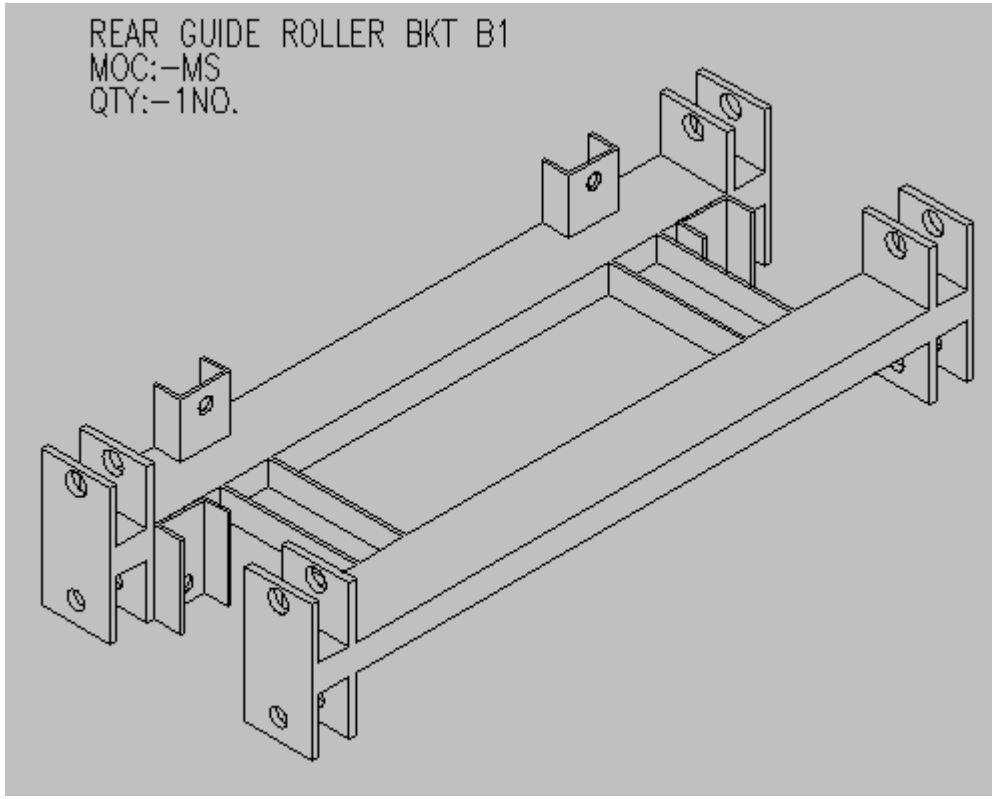
TELESCOPIC CONVEYOR

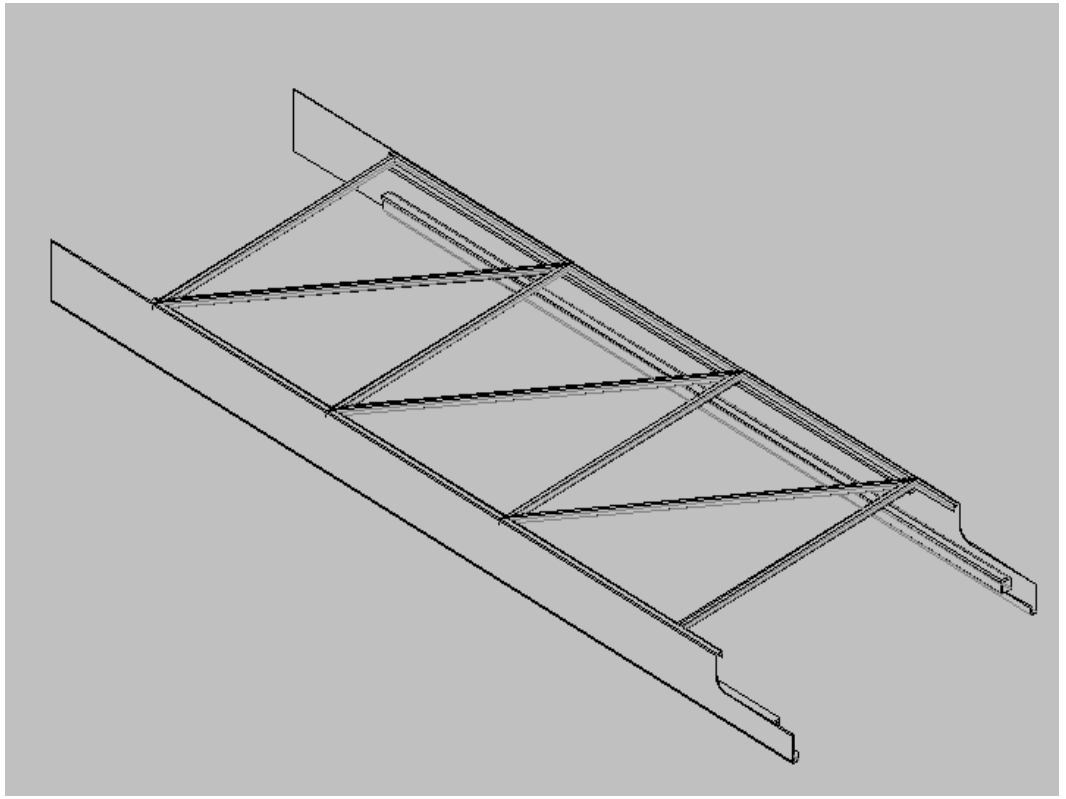


FRONTIER SCOPE:-

- ✓ Design, Manufacturing, Assembly
- ✓ Base Structure
- ✓ Rollers
- ✓ Bearings
- ✓ Sprockets & Chains
- ✓ Belts Food Grade
- ✓ Motor & Drive Packs
- ✓ Anchoring Unit
- ✓ Electronics (PLC, 1 HMI unit.)
- ✓ Mechanical Stopper
- ✓ Tower Lamp
- ✓ Photo Sensor
- ✓ UPS 20 min backup
- ✓ Hydraulic cylinder & power pack FA Std (achive/delta hyd)
- ✓ Head Pulley Guard
- ✓ Chain Guard
- ✓ Hardware
- ✓ Part Trails & Proofing

Basic Boom Structure





Conveyor Main Drive

2 Decision of any condition for input & output

① Motor speed	N =	1500 ▼	r/min

3 Decision of conveyance speed and reduction ratio

✔ In the case that conveyance speed is deci | In the case that redcution ratio is decided

① Speed	V =	30	m/min
②Reduction ratio (calculated value at ①)	Z =	39.27	

4 Caluclation of lead torque

4.1 Input of efficiency

① Efficiency of reducer and gearmotor	η_g =	0.79	
② Efficiency of the sprocket wheel or the gear	η_s =	1	

4.2 Calculation result (load torque and motor power)

③ Load tprque (Converted in a motor shaft)	Tf =	16.6012	N·m
④Conversion from load torque Tf to motor-power Pf	Pf =	2.6075	kW
④ Max. torque working on reducer (on output shaft)	To =	1708.232	N·m

Gear motor Specifications:-

- ✓ Motor Power :- 2.6 Kw or above
- ✓ Motor Shaft Speed: 1440 rpm
- ✓ Torque :- 1708 Nm or above
- ✓ Gear Ratio : 30
- ✓ Motor Poles :- 4
- ✓ 3 phase motor 415 volt 50 Hz
- ✓ Type: Helical worm
- ✓ Make: SEW (Customer Specified)

Note: above Selection may vary during final design process.

Conveyor Boom Drive

2 Decision of any condition for input & output

① Motor speed	N =	1500 ▼	r/min

3 Decision of conveyance speed and reduction ratio

✔ In the case that conveyance speed is deci | In the case that redcution ratio is decided

① Speed	V =	15	m/min
②Reduction ratio (calculated value at ①)	Z =	62.83	

4 Caluclation of lead torque

4.1 Input of efficiency

① Efficiency of reducer and gearmotor	η_g =	0.79	
② Efficiency of the sprocket wheel or the gear	η_s =	1	

4.2 Calculation result (load torque and motor power)

③ Load tprque (Converted in a motor shaft)	Tf =	3.5575	N·m
④Conversion from load torque Tf to motor-power Pf	Pf =	0.5588	kW
④ Max. torque working on reducer (on output shaft)	To =	871.453	N·m

Gear motor Specifications:-

- ✓ Motor Power :- 0.56 Kw or above
- ✓ Motor Shaft Speed: 1440 rpm
- ✓ Torque :- 871 Nm or above
- ✓ Gear Ratio : 63
- ✓ Motor Poles :- 4
- ✓ 3 phase motor 415 volt 50 Hz
- ✓ Type: Helical worm
- ✓ Make: SEW (Customer Specified)

Note: above Selection may vary during final design process.

Customer Scope of Supply

- ✓ Process Plan, process flow charts, WI and other documents related to Production
- ✓ Participation/DAP Approval for Designed equipment's before start of manufacturing
- ✓ 2D/ 3D Drawings of components.
- ✓ Trial Components, to and fro transportation, testing equipment's
- ✓ Unloading, Unpacking and Site placement
- ✓ Handling Equipment's for installation site.
- ✓ Electrical connection at necessary location
- ✓ Provision of primary air connection at all locations [If required]
- ✓ Site for installation (Civil work if any after submission of drawing by maker)
- ✓ Structural requirement and support (if any) will be made available by you at your cost.
- ✓ All the necessary consumables at FA and at Customer site during trials.

Commercial Offer -

Sr. No.	*Item Descriptions*	*Qty.*	*Rate*	*Amount*
1	Telescopic Conveyor 10.8 meter(Angle Adjustment Unit & HMI Unit)	1	**1,785,240**	**1,785,240**

Payment Terms

Prices are Ex –Works Frontier Automation, Pune.

- ✓ All the taxes and duties will be extra as applicable at the time of material delivery.
- ✓ Packing will be extra @ 2% on your account will be add in final dispatch invoice.
- ✓ Insurance will be on your account.
- ✓ Transportation charges from Frontier Automation to your site will be on your account, at actual.

Payment Terms

- ✓ 50% advance along with techno-commercially clear Purchase Order.
- ✓ 40 % with all the taxes and duties against Performa invoice **before dispatch** of material from FA.
- ✓ 10 % After Installation & commissioning

Warranty

- ✓ 12 months

Delivery

- ✓ 8-10 weeks from the receipt of advance.

Validity

- ✓ 30 days from the date of this commercial offer.

Thank you

Frontier
AUTOMATION

Frontier Robotics & Automation Pvt Ltd

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Chakan Phase - 1, Kurali, Chakan, Pune 410 501

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